

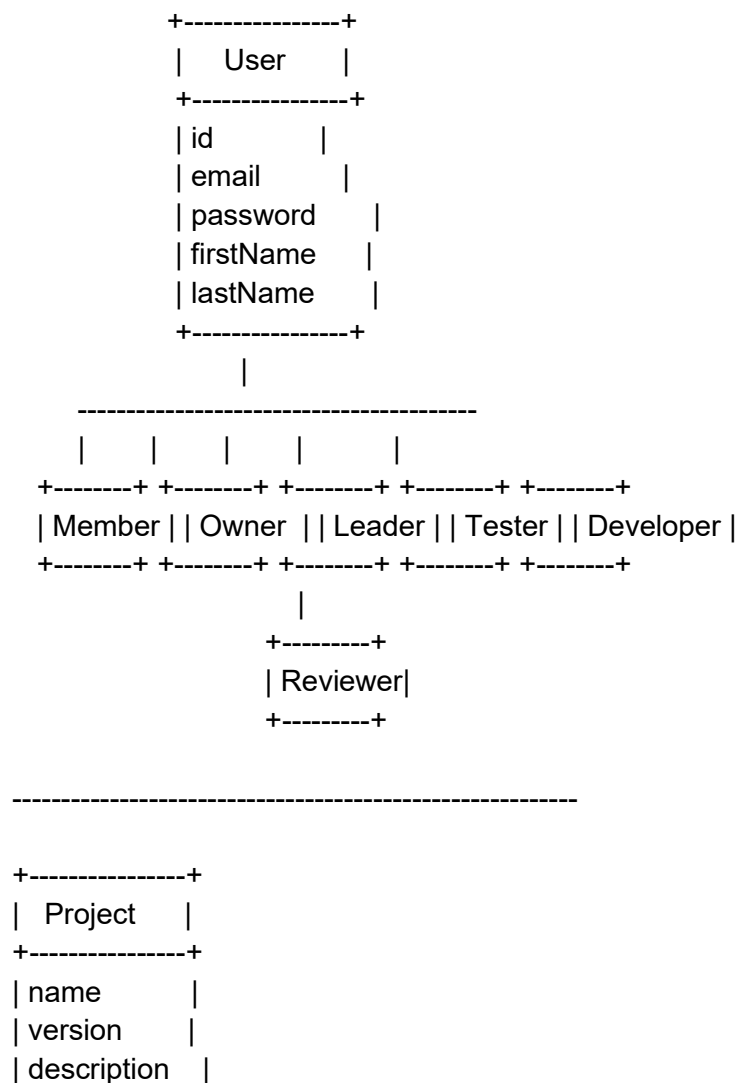
UML Class Diagram with AOP integration include:

- Core domain classes
 - Role hierarchy
 - Relationships
 - AOP Aspects integration layer
-

1. UML Class Diagram (Textual Representation)

You can directly convert this into **StarUML** / **Draw.io** / **PlantUML**.

Main Classes + Relationships



+-----+	
	1..*
+-----+	
	Invitation
+-----+	
	email
	role
	status
	link
+-----+	

+-----+	
	Pipeline
+-----+	
	status
	phase
+-----+	
	1..*
+-----+	
	Update
+-----+	
	title
	description
	versionType
	status
+-----+	
	1..*
+-----+	
	File
+-----+	
	fileName
	fileType
+-----+	

+-----+	
	Task
+-----+	
	title

```
| priority  |
| status   |
+-----+
```

```
+-----+
|  Bug    |
+-----+
| description |
| severity  |
| status    |
+-----+
```

(Update) ---- links ----> Task
(Update) ---- links ----> Bug

```
+-----+
| CodeReview |
+-----+
| notes      |
| decision   |
+-----+
```

```
+-----+
| TestCase  |
+-----+
| name      |
| result    |
+-----+
```

(Update) --> CodeReview
(Update) --> TestCase

2. Relationships Summary

Inheritance (Generalization)

User <|-- Member
User <|-- Owner
User <|-- Leader
User <|-- Tester
User <|-- Developer
User <|-- Reviewer

Associations

- Project → Invitations (1..*)
 - Pipeline → Updates (1..*)
 - Update → Files (1..*)
 - Update → Task / Bug (many-to-many)
 - Update → TestCase (1..*)
 - Update → CodeReview (1..1 or 1..*)
-

3. AOP Integration Layer (IMPORTANT)

AOP is **NOT** a normal class relationship → it is shown as a **cross-cutting layer**.

AOP Package

```
<<Aspect>> LoggingAspect  
<<Aspect>> SecurityAspect  
<<Aspect>> ValidationAspect  
<<Aspect>> PipelineAspect  
<<Aspect>> NotificationAspect  
<<Aspect>> ExceptionAspect
```

AOP Connections (Cross-Cutting)

LoggingAspect -----> ALL Services

SecurityAspect -----> Owner, Leader, Developer, Tester, Reviewer

ValidationAspect -----> User, Invitation, Task, Bug, Update

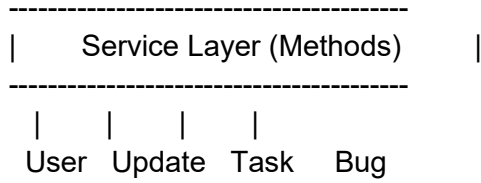
PipelineAspect -----> Pipeline, Update, CodeReview, TestCase

NotificationAspect ----> Invitation, Task, Bug, Update

ExceptionAspect -----> ALL Services

4. UML with AOP (Visual Concept)

<<Aspect>>
LoggingAspect
|
v



<<Aspect>>
SecurityAspect
|
v
Owner / Leader / Developer / Tester

<<Aspect>>
PipelineAspect
|
v
Pipeline ---> Update ---> TestCase
|
CodeReview
